Site Analysis & Human-Centric Research

What the site is, who it serves, and what is wrong with it

A flat, largely exposed 15,000 m² neighbourhood park serving roughly 7,640 residents within a ten-minute walk. Its defining problem is not layout or planting: for 55.5% of daylight hours, standing outside on it is uncomfortable or worse.

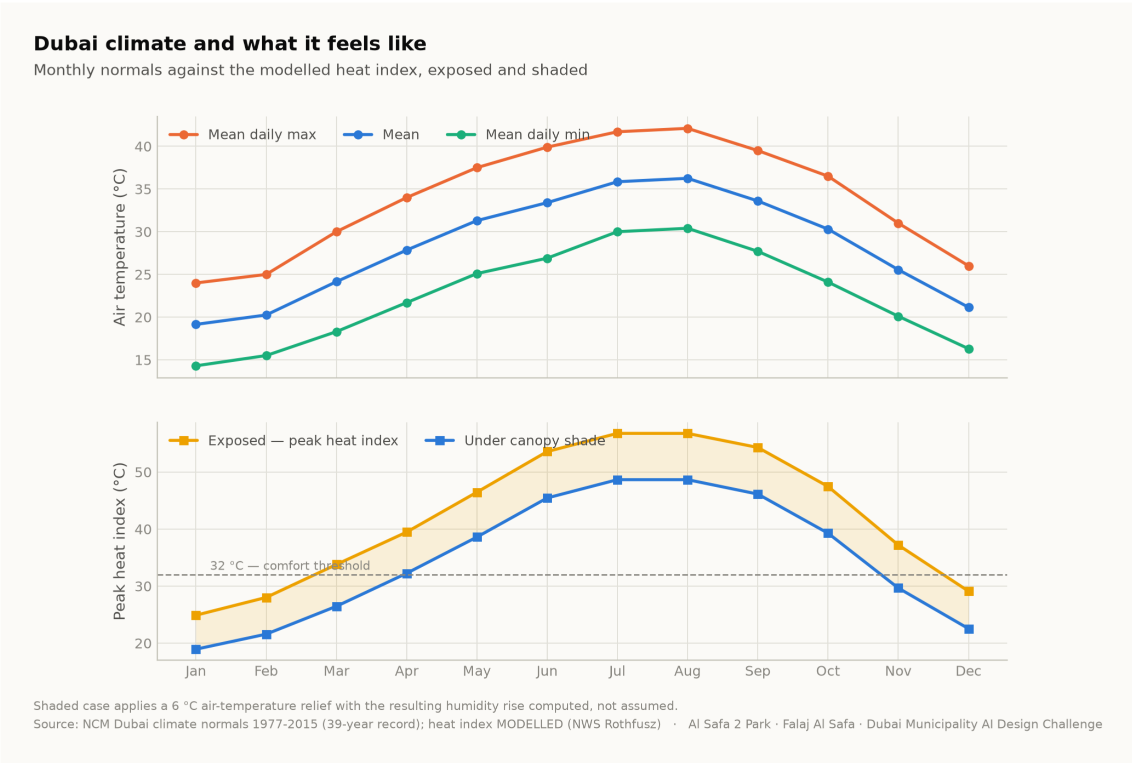
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. Climate — the governing constraint

The analysis reconstructs an 8,760-hour year for the site from 39 years of National Centre of Meteorology monthly normals, with solar positions computed for every hour by the NREL algorithm at the site's coordinates. The reconstruction is verified back against the published normals to within 0.39 °C.

The climate series is modelled, not measured. The underlying record is 39 years long but is published as monthly normals; the hourly series is reconstructed from them and labelled as modelled everywhere it appears. Conclusions are about a typical year, never about extremes.



Dubai climate normals against the modelled heat index. Analysis output — computed from project data.

2. The problem, ranked rather than listed

Problem	Severity	Evidence
Thermal discomfort	Critical	Only 44.5% of 4,402 daylight hours are comfortable; peak heat index 56.8 °C
No continuous shaded route	Critical	A straight or fragmented shade strategy leaves 330 hours a year with no shade anywhere along the primary route
Exposed perimeter	High	Roads on the boundary contribute noise, glare and reflected heat with no intervening mass
Water and irrigation demand	High	Open water or irrigated turf in this climate carries a continuing evaporation and supply cost
Limited programme range	Moderate	Without comfort, no amount of programming produces use

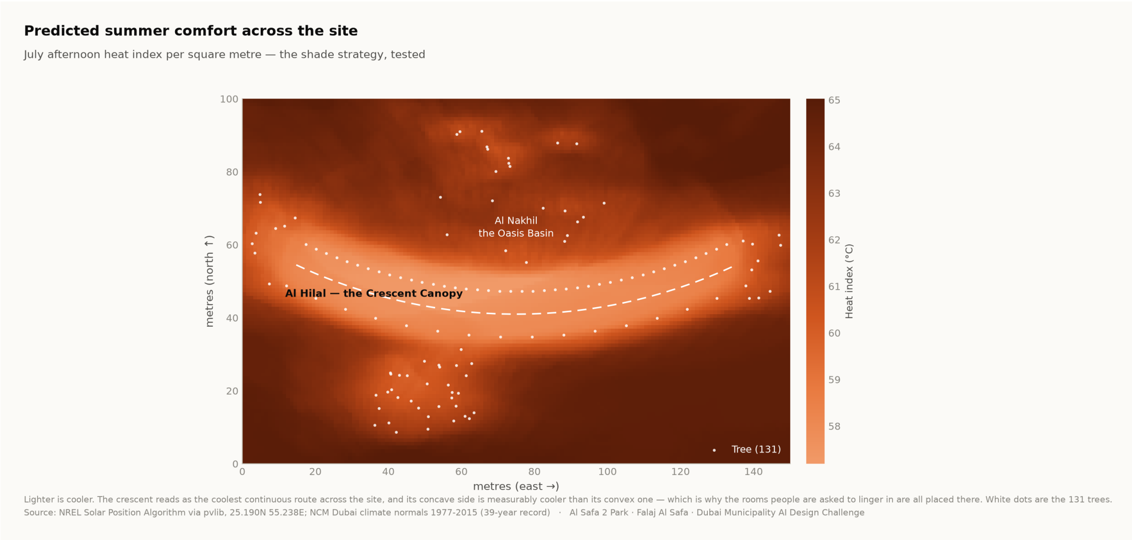
Ranking by severity is what justifies making shade the organising idea rather than one strategy among several.

3. Who the park serves

Approximately 7,640 residents live within a ten-minute walk (Dubai Statistics Center, 2023). The neighbourhood is established and low-rise, with families, working adults, domestic staff and a significant older population. The user groups and their needs are set out in the User Experience and Activation Strategy.

Visitor demand is a scenario, not a prediction. No footfall data exists for this site. Demand figures are deliberately excluded from the machine learning suite, because a model trained on an assumed demand curve would only recover the assumption that produced it.

4. Where the site is comfortable, square metre by square metre



Predicted July afternoon heat index per m². The crescent reads as the coolest continuous route, and its concave side is measurably cooler than its convex face. Analysis output — computed from project data.

5. Opportunities the analysis identifies

- **One continuous shaded route is worth more than dispersed shade.** Permutation importance puts distance to the crescent far above every other geometric feature.
- **Comfort is predictable from the calendar.** At 97.5% accuracy from sun position and date alone, park operations need no sensor network.
- **The shoulder seasons are recoverable.** The comfort gain concentrates in late afternoon in spring and autumn — hours that are currently lost and are cheap to win back.
- **The perimeter is an asset.** A planted berm addresses noise, glare and heat simultaneously.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.